

VGGNet Quiz – Conceptual Understanding

EASY LEVEL

Q1. Why is VGGNet considered a “deep” network?

- A) Because it uses large images
- B) Because it contains many sequential layers
- C) Because it removes pooling operations
- D) Because it only uses fully connected layers

Answer: B

Q2. What is the main purpose of convolution filters in VGGNet?

- A) Compress image files
- B) Detect patterns such as edges and textures
- C) Store class labels
- D) Normalize image colors

Answer: B

Q3. Why does VGGNet mainly use small convolution filters?

- A) To completely remove overfitting
- B) To increase image resolution
- C) To build deeper feature extraction efficiently
- D) To avoid using activation functions

Answer: C

Q4. What is the main role of Max Pooling in VGGNet?

- A) Increasing feature map size
- B) Keeping only the most important information
- C) Replacing convolution layers
- D) Increasing the number of parameters

Answer: B

Q5. In the first layers of VGGNet, the network usually learns:

- A) Full object categories
- B) Dataset labels
- C) Simple visual patterns like edges
- D) Final classification scores

Answer: C

MEDIUM LEVEL

Q6. Why are several small convolutions often better than one large convolution?

- A) They add more non-linear learning stages through repeated ReLU activations
- B) They remove the need for training data
- C) They eliminate pooling operations
- D) They always make training faster

Answer: A

Q7. What happens to learned features as we move deeper into VGGNet?

- A) Features become simpler
- B) Features become more abstract and meaningful
- C) Features disappear after pooling
- D) Features stop changing

Answer: B

Q8. What does it mean when we say the receptive field increases in deeper layers?

- A) The network forgets earlier features
- B) Neurons can understand larger regions of the image
- C) Pooling layers become larger
- D) The image resolution increases

Answer: B

Q9. Why is ReLU important in VGGNet?

- A) It increases image size
- B) It introduces non-linearity, helping the network learn complex patterns
- C) It replaces convolution operations
- D) It reduces the number of filters

Answer: B

Q10. Compared to AlexNet, what was VGGNet mainly trying to improve?

- A) Using shallower architectures
- B) Using fewer convolution layers
- C) Learning richer features using deeper networks and smaller filters
- D) Removing fully connected layers completely

Answer: C

HARD LEVEL

Q11. Why can deeper VGG layers recognize complex objects while early layers cannot?

- A) Because deeper layers combine simpler features learned earlier
- B) Because pooling layers perform classification
- C) Because early layers do not use filters
- D) Because later layers stop using convolution

Answer: A

Q12. Why does VGGNet require high computational resources?

- A) Because deep architectures perform many convolution operations
- B) Because pooling layers are computationally expensive
- C) Because ReLU stores all image pixels
- D) Because VGGNet avoids feature sharing

Answer: A

Q13. Why are pre-trained VGG models useful for transfer learning?

- A) Because they already learned useful visual representations from large datasets
- B) Because they do not require training anymore
- C) Because they work only on ImageNet
- D) Because they eliminate feature extraction

Answer: A

Q14. What problem appears when networks become very deep, leading to architectures like ResNet?

- A) Feature maps become colorful
- B) Gradients become difficult to propagate through many layers
- C) Pooling layers disappear
- D) Convolution filters stop learning edges

Answer: B

Q15. Which statement best describes the learning strategy of VGGNet?

- A) Learning through repeated small-step feature extraction across many layers
- B) Learning using only one large convolution layer
- C) Learning without activation functions
- D) Learning mainly through pooling operations

Answer: A

Test yourself

You can test yourself using this link as it has the same questions

<https://stirring-unicorn-9d416d.netlify.app/>